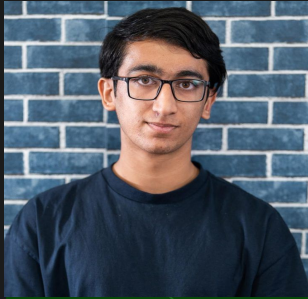




DE Capstone WatchFall

Meet the Team



Ibrahim Khurram
Software Engineer



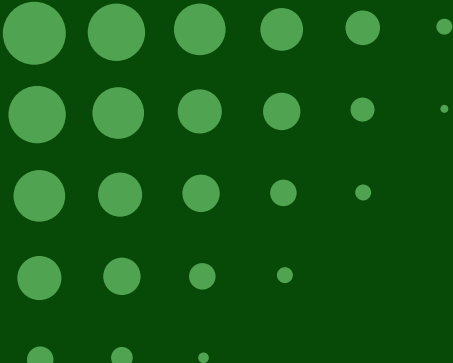
Nicolas Berger
Systems Engineer



Abdullah Khaled
Hardware Engineer



Caleb Rasband
Communication Engineer

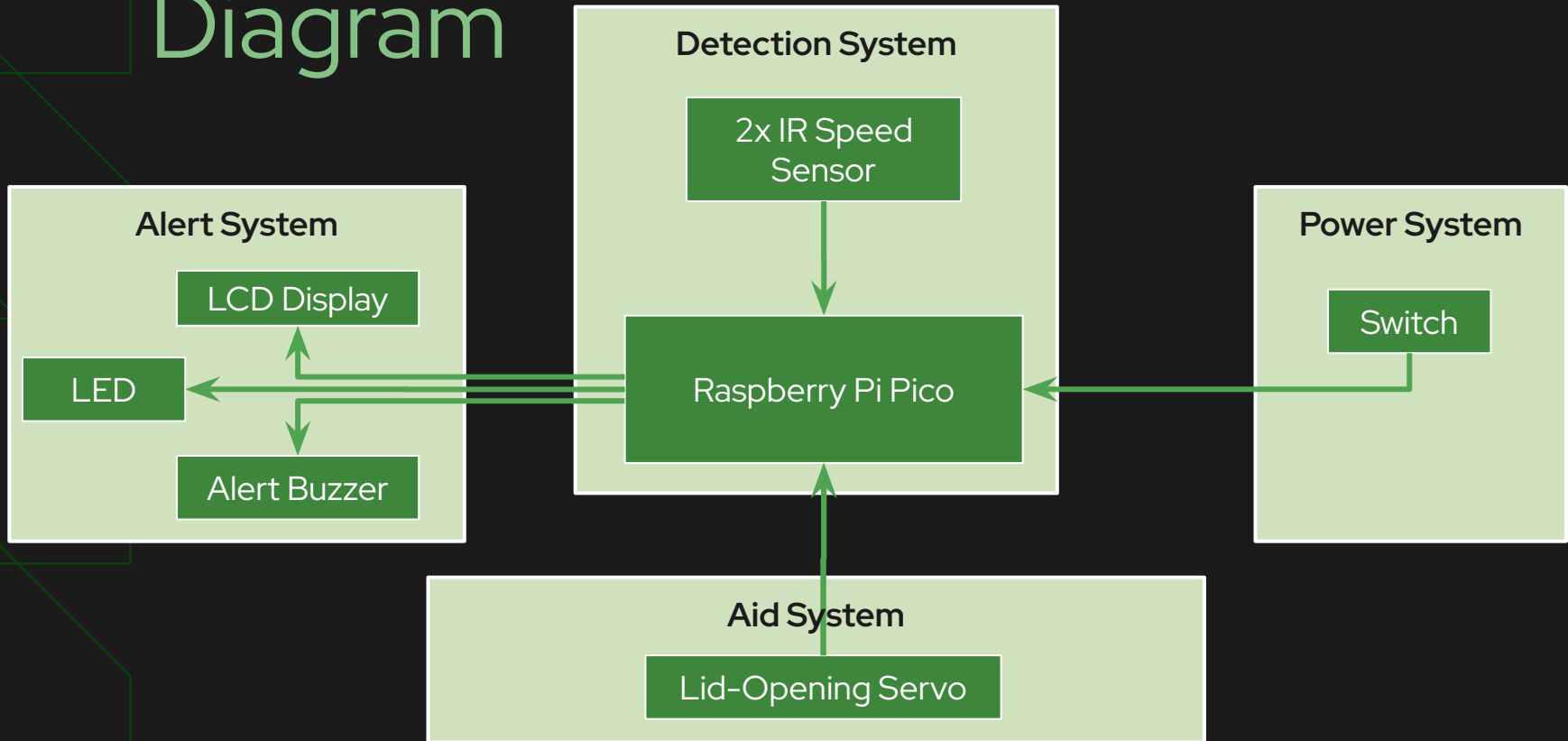


The Problem

Senior citizens aren't always able to care for themselves. Simple events like falling down can be big challenges, and help isn't always immediately available.

Our project addresses this by flagging seniors who go through these issues, covering a critical gap in safety.

Circuit Block Diagram





How it Works



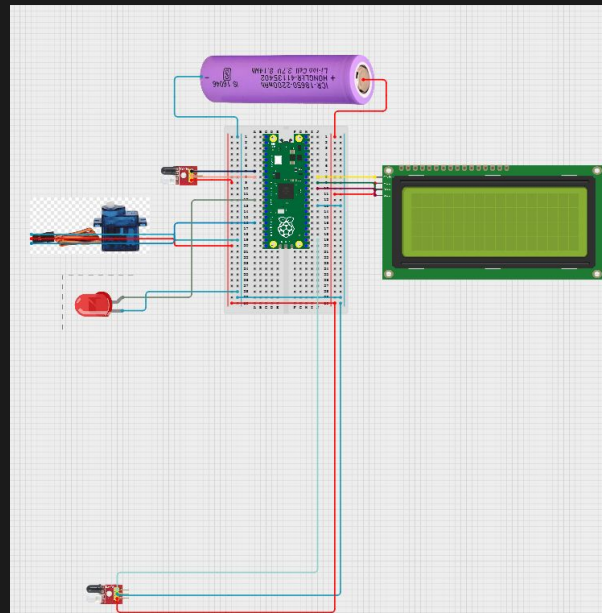
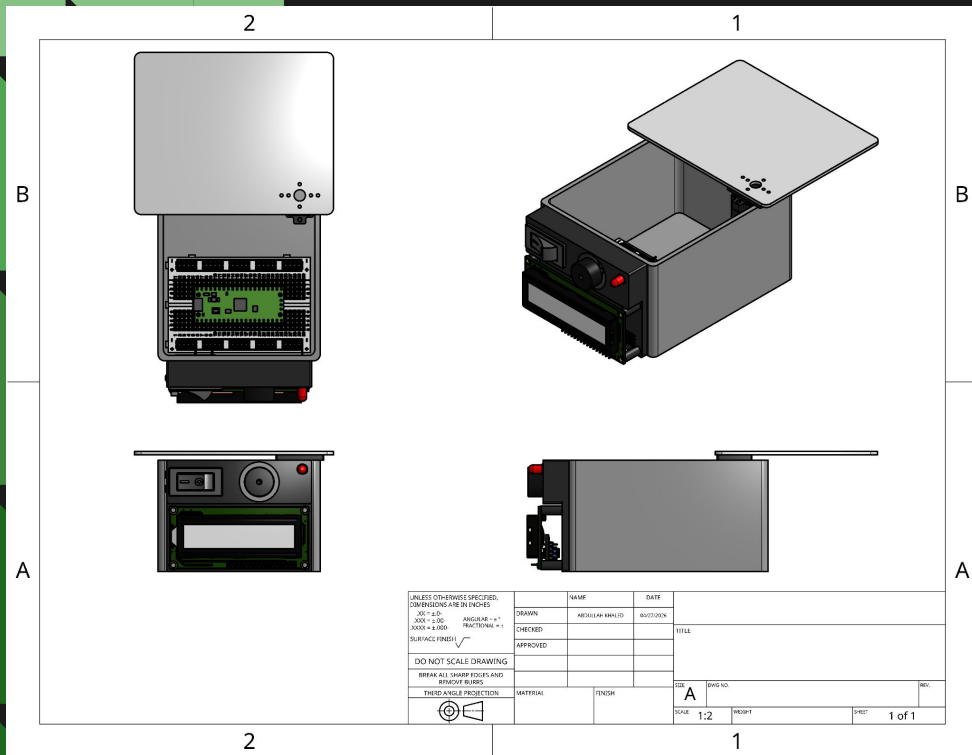
At a high level, WatchFall works by calculating the downwards velocity of the senior citizen and determining whether they are falling or not. There are two speed sensors, spaced at a measurable distance apart, which use infrared light to perform object detection. Once the senior is not detected on the first sensor, then a timer is started, which runs until the senior is detected by the second sensor. Then, the program can calculate the velocity by dividing the distance between the two sensors by the time it took for the senior to move between them. The program can then determine if the senior is falling using the threshold the program has for a “falling velocity,” with an understanding of the distance between the two sensors. If the senior is indeed falling, then the program will simultaneously flash an LED, play a buzzing sound, and open a servo door with medication the senior caretaker can administer to the senior (such as painkillers and bandages). Finally, after a certain amount of time, the device is set back to its initial state and is primed to detect another fall.

Inputs: Speed Sensors

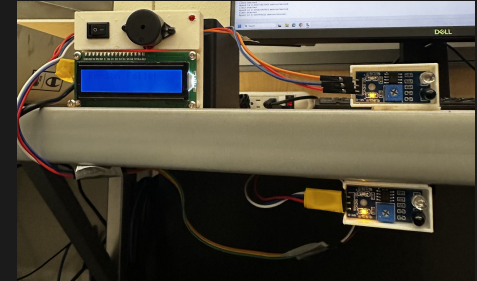
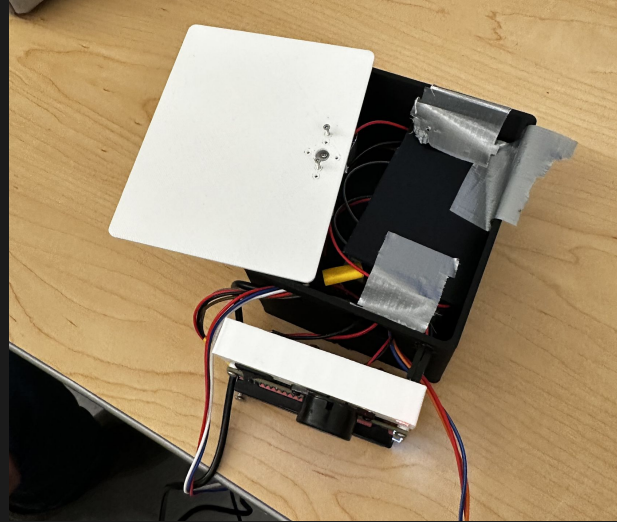
Processing: Velocity Calculation, Falling Determination

Output: LCD, Buzzer, Servo Lid

Images



Demo Images



What Did We Learn

One important thing that we learned was to control our scope. At the start of the project, we wanted to create an ML-powered smart monitor to determine if a senior had fallen anywhere in the room. However, we found that with the hardware constraints of the project (i.e. Raspberry Pi Pico H), it was infeasible. Therefore, we learned how to adjust our goals to provide a high-quality product that still fit within the constraints of our components.